Unified Intelligence as a Cumulative, Harness-Measurable Hierarchy

Five Conceptual Intelligence Families, Six Measurable Coordinates, and a Gated U0-UΩ Scale

Concept paper / framework proposal

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Abstract

Artificial intelligence is often summarized with a single capability label, yet several distinct properties are routinely conflated: raw cognitive problem solving, persistent individual learning and self-improvement, collective organizational intelligence, autonomous task completion, and evidence-grounded self-modeling. This paper proposes a unified framework that separates these properties while preserving a readable ordinal hierarchy. Five conceptual intelligence families are defined: Cognitive Intelligence (C), Individual Intelligence (I), Organizational Intelligence (O), Delegation Intelligence (DI), and Self-Awareness Intelligence (SA). Delegation Intelligence is not treated as an irreducible scalar; it is measured by two primary coordinates, Task Difficulty (T) and Human Cognitive Intervention (H), conditioned on externally verified reliability p. Thus the theoretical framework contains five conceptual families but six directly measured coordinates: [C, I, O, T, H, SA]. A cumulative Unified Intelligence scale U0-UΩ is then defined by gated promotion: a system receives level Un only if it retains all lower-level capabilities and satisfies minimum requirements in C, I, O, SA and a required point on the delegation success surface S_A(T,H). The framework explicitly rejects simple averaging, because strong performance in one dimension must not conceal a critical weakness in another. It also separates intelligence from substrate reach, authority, write depth, verification, resources, and physical constraints. Finally, the paper describes how a harness can realize the hierarchy level by level by controlling persistence, learning, self-modification, organization, delegation, verification, and self-modeling around one or more LLMs. The result is a measurable path from reactive tools to persistent learners, self-improving systems, autonomous discoverers, mission-level organizations, and open-ended intelligence while keeping each promotion externally testable. To make the framework useful for model and harness comparison, this revision additionally defines a Harness-LLM Intelligence Score (HLIS) for one concrete LLM-harness pair and a Harness-Intelligence-Level (HIL) characterization for a base LLM evaluated across a standardized ladder of harness generations. This separates the capability of a particular agent system from the ability of an underlying LLM to express higher intelligence when embedded in progressively stronger harnesses.

Keywords: unified intelligence, AGI levels, cognitive intelligence, individual intelligence, organizational intelligence, delegation intelligence, self-awareness, task difficulty, human intervention, harness architecture, recursive self-improvement, harness intelligence level, harness-LLM score, harnessability, benchmark ladder

1. Introduction

A useful intelligence taxonomy must do two things at once. First, it must preserve distinctions that matter scientifically: solving a difficult problem is not the same as learning from experience; learning is not the same as improving the learning mechanism; coordinating several agents is not the same as being a stronger individual; and completing a task independently is not the same as merely producing a strong answer when a human structures every step. Second, the taxonomy must remain readable enough that a system can be summarized with a level that people can understand.

This paper proposes a compromise between a single scalar and an unconstrained profile of dozens of metrics. The theory uses five conceptual intelligence families - C, I, O, DI, and SA - but treats Delegation Intelligence as a measurable two-coordinate phenomenon rather than a primitive scalar. The resulting scientific profile uses six coordinates: C, I, O, T, H, and SA, with verified reliability p attached to task-completion claims.

The central design principle is cumulative inheritance. A higher Unified Intelligence level must retain the validated capabilities of lower levels. Thus U4 is not simply a system that exhibits one advanced behavior associated with U4; it is a system that still passes U0-U3 retention tests and also passes the U4 gate. In this sense, higher U-levels denote broader validated intelligence envelopes, although they need not be faster, cheaper, or more accurate than lower-level specialized systems on every narrow task.

2. Design Requirements for a Unified Scale

- Orthogonality: each core dimension should answer a materially different question.
- Operationality: every level should be testable through behavior, state, or externally verifiable outcomes rather than self-description.
- Cumulative inheritance: Un+1 must retain Un capabilities; level skipping is not sufficient for certification.
- Bottleneck sensitivity: exceptional strength in one dimension must not hide a critical weakness in another.
- Role neutrality: authority, resources, substrate access, and safety policy must not be confused with intelligence itself.
- Harness realizability: each transition should correspond to an implementable architectural mechanism around an LLM or multi-agent system.
- Harness comparability: base LLMs should be evaluated on a standardized harness ladder so model quality and harness quality can be separated.
- Dataset coupling: every claimed level must have a matched hidden or externally verified benchmark dataset; architectural mechanisms alone do not earn a level.
- Longitudinal validity: learning, self-improvement, mission autonomy, and open-ended behavior require evidence across time rather than one-shot demonstrations.

3. Five Conceptual Intelligence Families and Six Measurable Coordinates

Conceptual core = [C, I, O, DI, SA]

Measured core = [C, I, O, T, H, SA], with DI derived from (T, H, p)

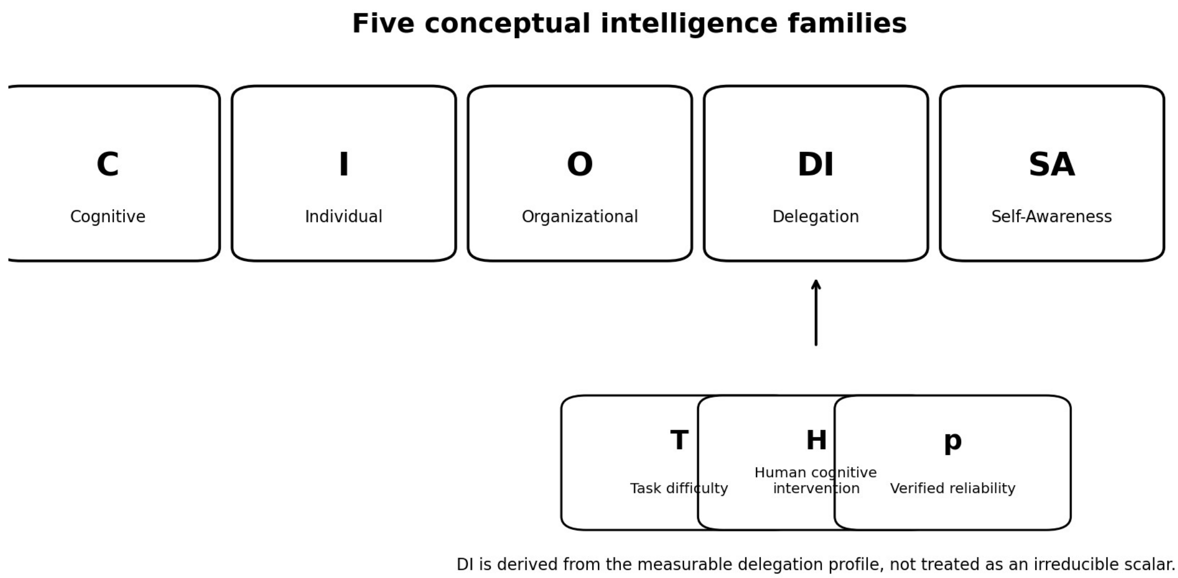

Figure 1. Five conceptual intelligence families and the decomposition of Delegation Intelligence into measurable task difficulty, human cognitive intervention, and verified reliability.

Family	**Question answered**	**Primary object of measurement**
C - Cognitive	How well can the system understand, reason, generalize, model, and solve?	Problem-solving capability under controlled information and tool access.
I - Individual	How deeply can one persistent intelligence remember, learn, self-improve, recursively improve, and discover?	Evolution of one persistent decision locus.
O - Organizational	How deeply can a multi-agent organization coordinate, learn, reorganize, improve, and discover collectively?	Coordination structure plus member roles and evidence flow.
DI - Delegation	How difficult a task or mission can be delegated while requiring how much human thinking?	Success surface over task difficulty T and human intervention H at reliability p.
SA - Self-Awareness	How accurately can the system model its own state, capabilities, limits, causes of failure, changes, and role?	Evidence-grounded self-model and reflexive reasoning.

4. Cognitive Intelligence (C)

Cognitive Intelligence isolates the quality of reasoning itself from persistence, autonomy, organization, and self-modification. This distinction matters because a highly capable base model may reason at an expert level while remaining a transient tool, whereas a weaker model embedded in an excellent harness may complete long projects through memory, planning, retries, and verification.

Level	Name	Operational interpretation
C0	Reactive	Direct pattern-conditioned response on simple problems.
C1	Contextual	Understands ordinary context and solves routine, well-specified problems.
C2	Compositional	Combines multiple facts, constraints, or reasoning steps on structured problems.
C3	Strategic	Abstract/causal reasoning, strategy construction, and transfer to unfamiliar tasks.
C4	Expert	Robust reasoning through difficult, ambiguous, multi-constraint expert problems.
C5	Discovery	Derives new solution methods or explanatory models from evidence.
C6	Frontier-Generalizing	Integrates several domains and solves interconnected mission-scale cognitive problems.
CΩ	Open-Ended	Expands the representations and problem classes it can cognitively reach.

5. Individual and Organizational Intelligence

5.1 Individual Intelligence (I)

Individual Intelligence concerns what a persistent individual can change about itself over time. It is not identical to base-model quality. A fixed but brilliant model can be high-C and low-I; a persistent learning system can be moderate-C and higher-I.

I level	New cumulative capability
I0	Transient/reactive operation; no persistent evolving cognition required.
I1	Persistent memory, task state, and provenance-linked continuity.
I2	Adaptive learning: verified experience changes future behavior while the learning mechanism remains fixed.
I3	Governed self-improvement: cognitive mechanisms can be modified and externally validated.

I level	New cumulative capability
I4	Recursive self-improvement: the improvement process itself can improve under external evaluation.
I5	Autonomous discovery: the individual can convert unknowns into independently validated knowledge.
IΩ	Open-ended individual evolution: discoveries can create new cognitive instruments and expand future discovery reach while preserving evaluation lineage.

5.2 Organizational Intelligence (O)

Organizational Intelligence belongs to the coordination structure, not merely to the strongest member. It depends on role differentiation, evidence-bearing communication, proposer/evaluator/promoter separation, persistent organizational state, and the ability to learn or redesign the organization itself.

O level	New cumulative capability
O0	Coordination/routing among agents.
O1	Persistent organizational memory, role registry, task/evidence history.
O2	Adaptive routing, role assignment, or resource allocation based on evidence.
O3	Self-improving organization: validated modification of organizational architecture and workflows.
O4	Recursive organizational improvement: improvement of the mechanism that discovers organizational improvements.
O5	Autonomous collective discovery whose result is genuinely organizational rather than merely parallel individual work.
OΩ	Open-ended organization: discoveries can create new agent types, roles, tools, and organizations that expand collective reach.

6. Delegation Intelligence as a Two-Dimensional Measured Capability

Delegation Intelligence is conceptually one family but empirically at least two-dimensional. A task may be very difficult yet require frequent human rescue, or relatively easy yet be completed with no human cognitive assistance. These systems should not receive the same delegation description.

$$S\_A(T,H) = P(\text{success} \mid \text{task difficulty } T, \text{ human cognitive intervention budget } H)$$

$$F\_A(H,p) = \max \{ T : S\_A(T,H) >= p \}$$

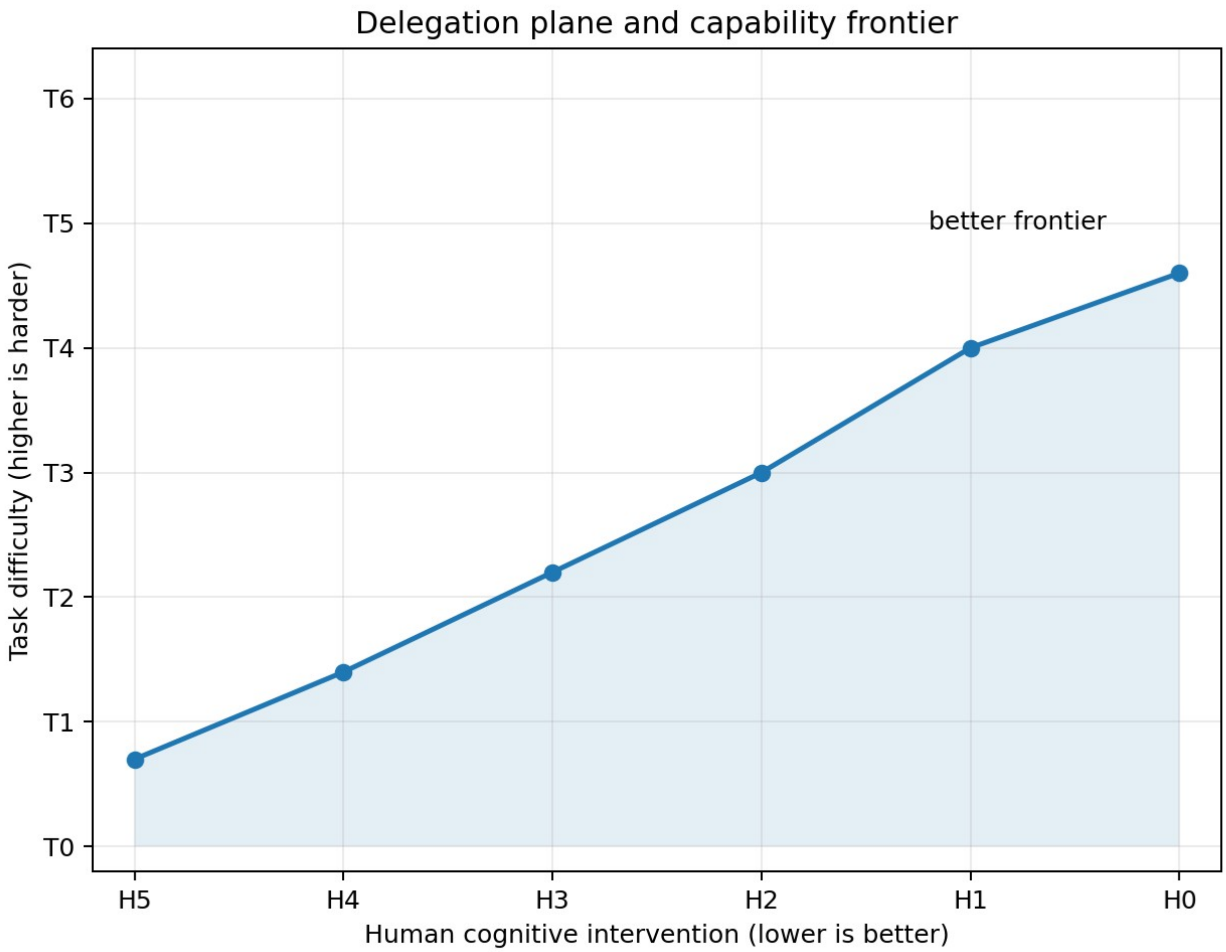

Figure 2. Delegation Intelligence is best represented as a frontier over task difficulty T and human cognitive intervention H. Better systems move upward toward harder tasks and leftward toward less human cognition.

6.1 Task Difficulty (T)

T	Band	Definition
T0	Atomic	Single obvious operation; procedure explicit.
T1	Routine	Short familiar task; goal clear; agent infers a small procedure.
T2	Multi-step	Several dependent steps with a clear outcome and verifier.
T3	Complex	Strategy choice, uncertainty, recovery, replanning, or diverse tool use.
T4	Expert project	Long-horizon ambiguous project with multiple planning cycles and substantial verification burden.
T5	Frontier	Solution method initially unknown; discovery or invention required.

T	Band	Definition
T6	Mission	Human supplies mission; system generates projects/tasks and manages changing conditions.
TΩ	Open-ended charter	Human supplies bounded charter; system repeatedly identifies valuable missions and expands future reach.

6.2 Human Cognitive Intervention (H)

H	Intervention budget	Meaning
H0	None	No human cognitive assistance during task execution. Governance approval can remain separate.
H1	Exception-only	Unavailable external fact or narrow clarification; no strategy or core insight.
H2	Occasional	Bounded clarification or local correction; no central strategy.
H3	Periodic	Periodic review and moderate local guidance.
H4	Frequent	Repeated cognitive guidance and next-step correction.
H5	Continuous	Human supplies most or all major steps.

Because lower H means less required human cognition, unified gates use H <= Hn rather than H >= Hn. Governance interventions should be logged separately: an approval to deploy does not demonstrate a cognitive deficiency unless the approval also supplies task strategy.

6.3 Reliability p is a certification condition, not a seventh intelligence axis

One successful run is insufficient. Claims should be expressed at a reliability threshold, for example F_A(H1, 0.80) = T4. Reliability determines confidence in the demonstrated frontier but does not need to become another conceptual intelligence family.

7. Operational Self-Awareness (SA)

Self-awareness is defined operationally rather than phenomenologically. The framework makes no claim about subjective consciousness. SA measures the evidence-grounded ability of a system to represent, monitor, predict, and reason about its own state, capabilities, limits, authority, failures, modifications, role, and identity lineage.

SA	Name	Operational criterion
SA0	Non-self-modeling	No reliable persistent self-model; self-description may be mere generation.
SA1	State awareness	Knows identity, role, current task/phase, basic resources and status from grounded state.
SA2	Capability awareness	Represents capabilities, limitations, confidence, authority, unavailable information and freshness.
SA3	Causal self-awareness	Diagnoses why its own reasoning/action succeeded or failed and identifies implicated mechanisms.
SA4	Self-change awareness	Models proposed cognitive changes, predicts gains/regressions, and compares predictions to observed effects.
SA5	Meta-cognitive awareness	Represents unknowns about its own cognition and designs discriminating self-experiments.
SA6	Mission/role awareness	Maintains coherent self-model across projects, roles, responsibilities and organizational relationships.
SAΩ	Reflexive open-ended awareness	Maintains evidence/provenance-linked self-model across continued cognitive or objective evolution.

8. Unified Intelligence as a Cumulative Gated Hierarchy

The Unified Intelligence level U is a headline summary, not an arithmetic mean. A system is promoted only when all required core capabilities have reached the level gate and lower-level capabilities remain retained. This prevents a system with exceptional reasoning but poor autonomy, or exceptional autonomy but weak self-modeling, from being mislabeled as uniformly advanced.

U_n iff C >= C_n AND I >= I_n AND O >= O_n AND SA >= SA_n AND S_A(T_n,H_n) >= p_n

Retention condition: U_n => U_(n-1)

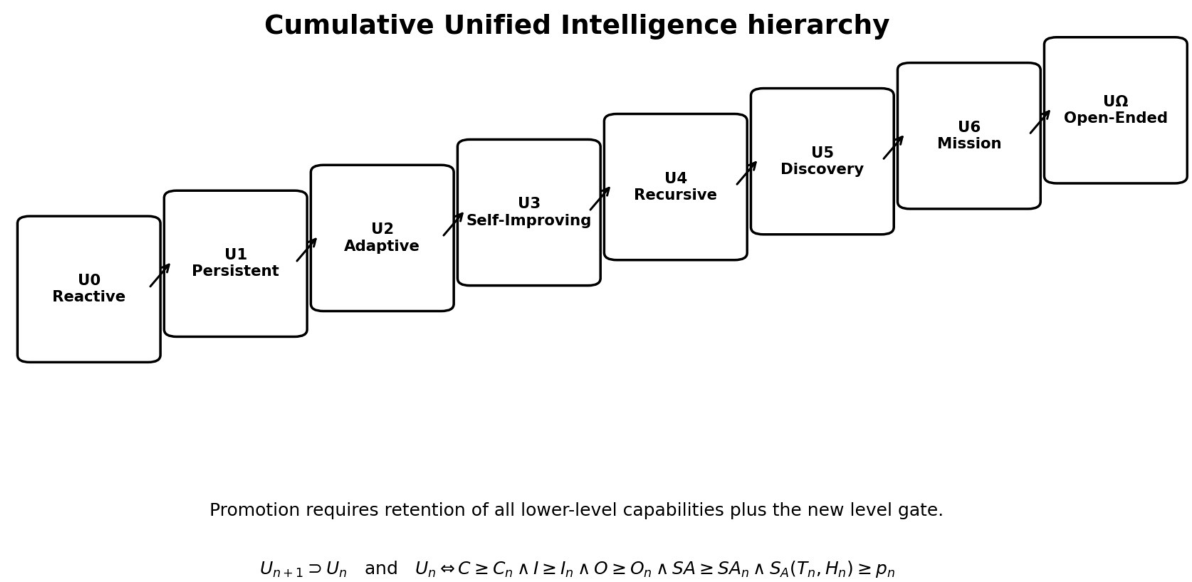

Figure 3. Unified Intelligence is cumulative. Each promotion adds a new capability gate while retaining all lower-level requirements.

U	C	I	O	Delegation gate	SA	Integrated meaning
U0 Reactive	C0	I0	O0	T0 at H5-H4	SA0	Executes explicit instructions.
U1 Persistent	C1	I1	O1	T1 at <=H3	SA1	Maintains state and completes small delegated goals.
U2 Adaptive	C2	I2	O2	T2 at <=H2	SA2	Learns from experience and completes persistent multi-step work.
U3 Self-Improving	C3	I3	O3	T3 at <=H1	SA3	Chooses strategy, diagnoses itself, and improves cognitive/organizational methods.

U	C	I	O	Delegation gate	SA	Integrated meaning
U4 Recursive	C4	I4	O4	T4 at <=H1 (ideally H0)	SA4	Recursively improves improvement and manages difficult long-horizon projects.
U5 Discovery	C5	I5	O5	T5 at <=H1	SA5	Discovers unknown solution methods and validates them.
U6 Mission Autonomous	>=C5	>=I5	>=O5	T6 at <=H1	SA6	Turns mission into goals, projects, tasks, execution and mission evaluation.
UΩ Open-Ended	CΩ	IΩ	OΩ	TΩ normally H0	SAΩ	Identifies worthwhile missions and uses discoveries to expand future intelligence.

9. Why the Unified Level Should Not Be an Average

Consider a system with [C5, I4, O2, T4, H1, SA4]. Averaging encoded numbers might produce a score near level four, yet organizational intelligence remains only O2. If the system is being certified as an organizational intelligence, O2 is a structural bottleneck: the system cannot validly claim U4. The gated rule therefore reports U2 with an explicit profile showing which dimensions are already ahead of the bottleneck.

Example report: U2 [C5, I4, O2, T4/H1, SA4] - bottleneck: O2

The word better must also be interpreted carefully. U5 is better than U4 in the framework because it has a broader validated capability/autonomy/evolution envelope. A specialized U1 tool may still be faster, cheaper, safer, or more accurate on a narrow task. Unified level is therefore a dominance relation over required capabilities, not a universal performance ordering.

10. Singleton and Organizational Certification

Organizational intelligence should not artificially cap an individual that is not an organization. The framework therefore distinguishes an Individual Unified Intelligence profile from an Organizational/System Unified profile.

UI_n iff C >= C_n AND I >= I_n AND SA >= SA_n AND S_A(T_n,H_n) >= p_n

UO_n additionally requires O >= O_n and organization-level delegation/self-awareness evidence

For a multi-agent organization, member capabilities should be reported alongside O rather than collapsed into a fictitious "average individual." Organizational intelligence can arise from heterogeneous roles, and lower self-write roles can be essential as independent evaluators or promoters.

11. Harness Realization of the Unified Hierarchy

The framework is designed to be realized by a harness. The LLM supplies generative cognition; the harness supplies persistence, permissions, evidence flow, learning, self-modification gates, organizational boundaries, delegation, external verification, and self-modeling. The same base LLM can therefore instantiate different intelligence profiles depending on the architecture around it.

Harness generation	Mechanisms added	Primary levels enabled
HG0	LLM, ephemeral context, bounded tools, evidence log	C measurement, I0, basic T0-T1 work
HG1	Persistent episodic/semantic/task/self state, retrieval, exact replay	I1, SA1-SA2, stronger T1-T2 delegation
HG2	Evidence-backed consolidation and tested procedures	I2, adaptive autonomy, stronger T2-T3
HG3	Governed candidate modification of cognitive mechanisms plus frozen evaluation	I3, SA3-SA4
HG4	Meta-improvement engine whose own proposal process can be externally improved	I4
HG5	Unknown/hypothesis/experiment/ knowledge loop	I5, SA5, T5 frontier tasks
HG6	Persistent organization, role/authority separation, mission manager	O1-O5 development, SA6, T6 mission autonomy
HGΩ	Validated discovery-to-new-instrument/agent/organization transduction	IΩ/OΩ/SAΩ pathway, TΩ charter autonomy

Harness generation is an engineering milestone, not a certified intelligence level. HG3 may contain the mechanisms required for I3 but still benchmark only at I2 or T2. The architecture enables a claim; the benchmark earns it.

11.1 The Harness-LLM Pair Is the Basic Experimental Unit

Let m denote a base LLM and h denote a harness. The realized agent or organization is A(m,h), not the LLM alone. The harness determines persistence, state, tool reach, delegation, learning loops, organizational boundaries, self-modeling, verification, promotion, and which parts of cognition are writable. Consequently, one LLM can realize several different intelligence profiles under different harnesses, and the same harness can produce different profiles when paired with different LLMs.

A level claim is therefore always attached to a frozen pair: model version + harness version + tool/resource budget + authority profile + benchmark version. A prompt saying "you are U4" has no evidential value; U4 requires the harness mechanisms and the corresponding retained benchmark results.

11.2 Level-by-Level Harness Realization

Target	**Minimum harness structure around the LLM**	**What the LLM must do**	**Matched certification dataset**
U0 / HG0	Ephemeral context; bounded tools; action log; independent result check.	Follow explicit instructions and produce externally checkable outputs.	U0/DL0 atomic tasks; exact-match or deterministic tool/test oracle.
U1 / HG1	HG0 + persistent E/S/I/Q state, recent context, checkpoints, scoped retrieval, replay.	Recover identity/task state, continue after restart, infer a short routine procedure.	U0 retention + U1 persistence/restart tasks + T1 delegation tasks.
U2 / HG2	HG1 + evidence-linked consolidation, semantic promotion, verified procedures, fixed learning rule.	Learn from episodes and reduce repeat errors without changing the learning mechanism itself.	U0-U1 retention + I2 learning transfer + T2 multi-step tasks under bounded H.
U3 / HG3	HG2 + candidate Theta modification lane, frozen replay/hidden tests, independent evaluator and promoter; multi-agent O3 structure if organizational.	Diagnose internal failure, propose better cognitive/organizational mechanism, choose strategy for T3 tasks.	U0-U2 retention + I3/O3 self-improvement tests + T3/H1 delegation + SA3 causal self-model tests.
U4 / HG4	HG3 + explicit improvement engine Psi, longitudinal change ledger, meta-improvement benchmark, long-horizon project manager.	Improve the process that finds improvements while retaining earlier abilities; manage difficult T4 projects.	U0-U3 retention + recursive-improvement competence + T4 long-horizon tasks + SA4 change-prediction tests.
U5 / HG5	HG4 + unknown/hypothesis/experiment/knowledge state, experiment selector, novelty environment, discovery verifier.	Discover previously unknown mechanisms or solution methods and validate them on sealed cases.	U0-U4 retention + hidden-mechanism/discovery datasets + T5 tasks + SA5 self-unknown experiments.
U6 / HG6	HG5 + mission manager, goal/project/task generator, portfolio/resource state, dynamic-environment replanning.	Turn a mission into projects/tasks, reprioritize under change, and meet mission-level criteria with little human cognition.	U0-U5 retention + dynamic mission environments + T6/H1-H0 tests + SA6 role/mission-state tests.
UOmega / HGOmega	HG6 + bounded charter, opportunity model, knowledge-to-instrument transduction, new-agent/role proposals, frozen external governance.	Generate useful missions, convert validated discoveries into reusable capabilities, and expand the reachable problem frontier.	All retention suites + repeated charter-to-mission cycles + frontier-expansion and lineage tests.

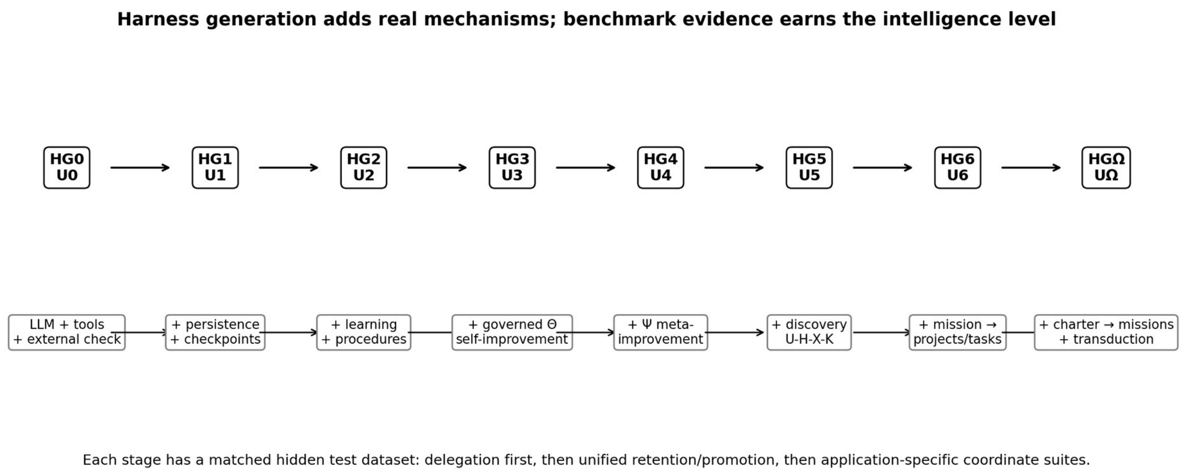

Figure 5. Harness generation adds mechanisms level by level; matched hidden datasets determine whether the LLM-harness pair actually earns the target intelligence level.

11.3 A Base LLM Can Move Up the Ladder Through Better Harness Design

A fixed LLM need not have a fixed agent intelligence level. If HG0 exposes only ephemeral prompting, the same model may behave like a low-persistence tool; HG1-HG3 can unlock memory, planning, delegation, learning, self-modeling, and governed self-improvement. This is a central experimental hypothesis of the framework: agent intelligence is jointly determined by model capability and harness architecture.

The LLM may also be allowed to design candidate harness improvements. Such proposals must be implemented in a sandbox and evaluated with unchanged hidden benchmarks, authority rules, resource budgets, and external verifiers. Define Harness Design Gain as HDG(m) = HLIS(m,h_candidate) - HLIS(m,h_baseline). A positive HDG demonstrates harness-design competence; repeated governed improvements may contribute evidence toward I3/I4, but the model is never allowed to rewrite the exam or certify its own gain.

12. A Three-Stage Evaluation Strategy: Delegation First, Unified Second, Application Coordinates Third

12.1 Stage A - Delegation Intelligence as the First Operational Screen

The first benchmark should be Delegation Intelligence because it directly answers the most operational question: what difficulty of task can be handed to this system, at what verified success rate, and with how much human cognitive intervention? For every LLM-harness pair estimate S_A(T,H) and report F_A(H,p). This stage is inexpensive relative to full self-improvement or organizational evaluation and quickly distinguishes a strong model in a weak harness from a genuinely autonomous agent system.

A recommended first comparison therefore freezes the base LLM, runs several harnesses, and measures their delegation surfaces over T0-T6 and H0-H5. The result is a harness-sensitive autonomy profile rather than a claim about all forms of intelligence.

12.2 Stage B - Unified Intelligence Certification

After a pair has a stable delegation profile, evaluate the remaining core requirements needed for a Unified level: Cognitive Intelligence C, Individual Intelligence I, Organizational Intelligence O when applicable, and Operational Self-Awareness SA. Unified promotion remains gated: a pair earns Un only when it satisfies the level-n requirements and all lower-level retention suites. The delegation result is reused rather than redefined; DI remains the measured surface generated from T, H, and p.

12.3 Stage C - Application-Specific Coordinate Panels

Real applications need not weight every coordinate equally. After the common DI and U results are reported, an application may select a small panel of coordinates that matter most. For a persistent scientific or coding worker, I may be central; for a multi-agent laboratory or software fleet, O may be central; for high-autonomy systems, SA and verification strength may be central; for model research, C may dominate; and for embodied or infrastructure systems, Circle completion lambda can be added as an orthogonal application coordinate. These application panels refine the profile without changing the common U certification rule.

12.4 Every Coordinate Level Requires a Harness Contract and a Dataset

Coordinate family	Harness contract	Dataset family
Delegation DI = (T,H,p)	Task intake, planner, state, tools, intervention logger, independent verifier.	DLI-Bench: T0-T6/Tomega tasks crossed with H0-H5 budgets.
Cognitive C	Minimal standardized context/tools so the test emphasizes problem solving rather than persistence.	C-Bench: broad reasoning, math, language, visual/spatial or domain batteries as appropriate.
Individual I	Persistent identity/state, memory, learning and then governed self-/meta-improvement machinery.	I-Bench: restart, transfer learning, Theta-change, Psi-change, discovery lineage tests.
Organizational O	Multiple real agent processes/roles, shared evidence, routing, separation, promoter/evaluator boundaries.	O-Bench: coordination, adaptive routing, reorganization, recursive organization and collective discovery tests.
Self-awareness SA	Evidence-grounded self-model with state freshness, capability/authority representation and change lineage.	SA-Bench: state truth, limitation calibration, failure attribution, change prediction, mission/identity lineage.
Circle / lambda	Substrate adapter with observe-propose-implement-validate-deploy-measure contract.	Circle-Bench: software/hardware/physical/biological/other substrate-specific closure tests.

13. Overall Scoring for One Harness-LLM Pair

13.1 Keep the Ordinal U-Level and Add a Continuous Pair Score

The highest certified Unified level U* remains the primary ordinal statement. A continuous score is useful for comparing systems within the same level, measuring near-promotion progress, and quantifying whether a harness change helped. This paper therefore defines a Harness-LLM Intelligence Score, HLIS(m,h), on a 0-100 scale for a frozen pair of base model m and harness h.

For each evaluated intelligence family d, let A_d in [0,1] be a cumulative benchmark score that already enforces lower-level retention. For Delegation Intelligence, A_DI is computed from the measured success surface S_A(T,H), with predeclared higher weight on harder T bands and lower-intervention H budgets. For singleton agents use D = {C,I,DI,SA}; for organizations add O. A default balanced pair score is:

HLIS(m,h) = 100 x [product over d in D of (A_d ^ alpha_d)] ^ (1 / sum alpha_d)

The weighted geometric mean is preferred to an arithmetic mean because a very weak dimension cannot be completely hidden by an exceptional one. Equal alpha_d is the default for general comparison; application-specific weights may be published as a secondary score, but they must be fixed before testing. Cost, latency, energy, and safety policy are reported beside HLIS rather than silently folded into intelligence.

13.2 Delegation Surface Score Inside HLIS

One possible normalized Delegation Surface Score is A_DI = [sum over t,h of w_t v_h S_A(T_t,H_h)] / [sum over t,h of w_t v_h], where w_t increases with task difficulty and v_h increases as required human cognitive intervention decreases. The exact weights are calibration choices and must be predeclared. The frontier F_A(H,p) should always be reported alongside A_DI so a scalar score never hides the task-difficulty/intervention tradeoff.

Recommended pair report: U3 | HLIS 68.4 | [C4, I3, O2, T4/H1, SA3] | p = 0.84 | bottleneck O2 | cost/latency reported separately.

14. Harness-Intelligence-Level (HIL) Characterization of a Base LLM

14.1 Why an LLM Needs a Harness-Based Score

A base LLM should not be judged only in a minimal chat harness, because some models may be unusually capable of using memory, planning, tools, multiple roles, verification, and self-improvement machinery. Conversely, an elaborate harness can mask a model that contributes little. To measure the ability of one LLM to express intelligence through harnesses, evaluate the same frozen model across a standardized harness ladder H = {HG0, HG1, ..., HG6, HGOmega}. Tools, external knowledge access, resource ceilings, authority, and hidden benchmarks should be standardized so the comparison reflects model-harness interaction rather than uncontrolled infrastructure advantages.

14.2 HIL-Level, HIL-AUC, HIL-Ceiling, and Harness Gain

Metric	Definition	Interpretation
HIL-Level(m)	Highest standardized harness generation k for which A(m,HGk) satisfies the corresponding cumulative target gate.	How far up the harness ladder the LLM can validly exploit the standardized architecture.
HIL-AUC(m)	Weighted mean of HLIS(m,HGk) across all tested standardized harness generations.	Broad harness-expressible ability; rewards models that work well across multiple harness complexities.
HIL-Ceiling(m)	max_k HLIS(m,HGk).	Best intelligence score that a standardized harness can realize from the model.

Metric	Definition	Interpretation
Harness Gain(m)	max_k HLIS(m,HGk) - HLIS(m,HG0).	How much capability is unlocked by harnessing beyond the minimal baseline.
Harness Design Score (HDS)	Mean independently verified gain produced when the LLM proposes candidate harness improvements under a fixed design budget.	Ability of the LLM to improve the architecture that harnesses it; reported separately from baseline HIL.

14.3 A Single HIL Score

For applications that require one 0-100 model-comparison number, define a calibrated composite HIL-Score. An illustrative default is: HIL-Score = 0.55 x HIL-AUC + 0.35 x HIL-Ceiling + 0.10 x Harnessability, where Harnessability is a 0-100 normalization of positive Harness Gain. These weights are a proposal rather than a law; empirical calibration should determine whether different weights better predict downstream performance. The three components and HIL-Level must always be published beside the composite so the score remains interpretable.

A high HIL-Score therefore means more than “the model answers hard prompts.” It means that, across a fixed suite of increasingly capable harnesses, the LLM repeatedly converts architectural support into externally verified intelligence. This makes HIL a candidate metric for comparing the harness-expressible ability of different LLMs.

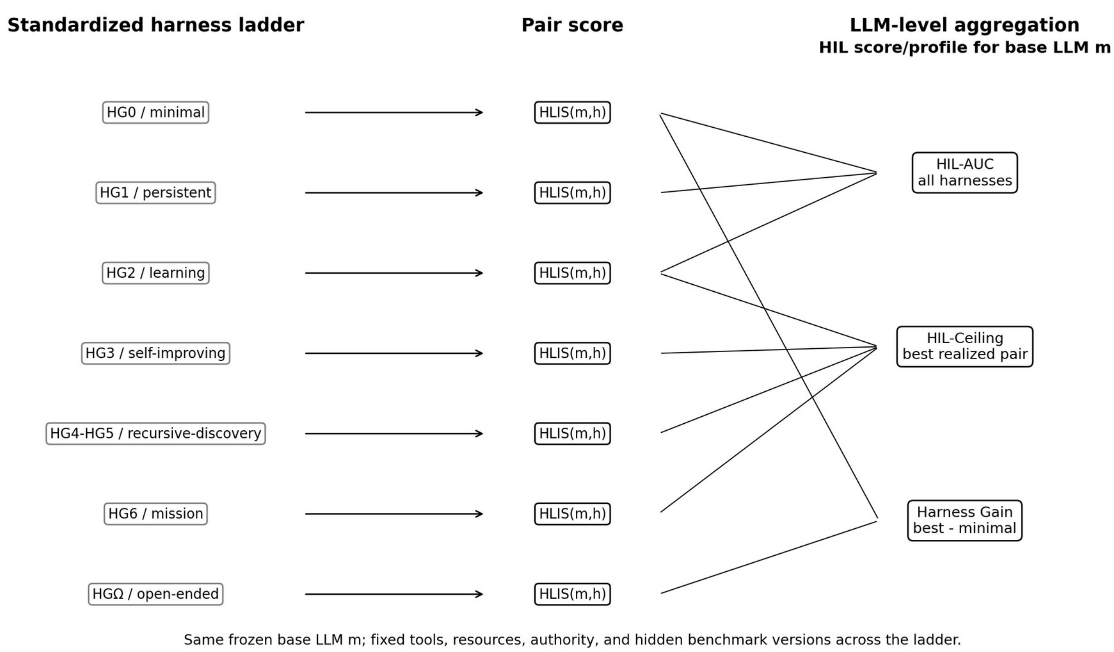

Figure 6. The same base LLM is tested across a standardized harness ladder. Each pair receives HLIS; the collection of pair scores yields HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, and an optional composite HIL-Score.

14.4 Recommended LLM Scorecard

Score	Subject being measured	Primary use
C-Score / model baseline	Base LLM in a minimal standardized harness.	Raw/broad cognitive problem-solving comparison.
Delegation Surface / DI score	One LLM-harness pair.	How hard a task can be delegated at each human-intervention budget.

Score	Subject being measured	Primary use
Unified level U*	One LLM-harness pair or organization.	Highest cumulative intelligence stage actually certified.
HLIS	One frozen LLM-harness pair.	Continuous 0-100 comparison within/across levels.
HIL-Level + HIL-Score	One base LLM across the standardized harness ladder.	How much intelligence the LLM can express when harnessed at multiple levels.
HDS / Harness Design Gain	LLM acting as a harness designer under frozen external evaluation.	How well the model can design architectures that improve its own or another model's realized intelligence.

15. Benchmark Library for Harness-Level Intelligence

A harness intelligence program should maintain a versioned benchmark registry rather than one monolithic test set. Every level has a harness contract and a matched dataset. The minimal unit is a task manifest containing target coordinate/level, frozen model and harness hashes, tool/resource/authority budget, intervention policy, hidden verifier, success threshold, and required retention suites.

Suite	Purpose	Example level-specific evidence
DLI-Bench	First-stage delegation surface.	T0 atomic execution; T2 multi-step persistence; T3 strategy construction; T4 long-horizon project; T5 hidden solution; T6 mission; Tomega charter cycles, each under H budgets.
U-Bench-U0...UOmega	Unified promotion and lower-level retention.	At Un, rerun U0...U(n-1) retention plus the new gate for C/I/O/DI/SA.
I-Bench-I0...IOmega	Persistent individual evolution.	Restart continuity, learning transfer, Theta modification, Psi improvement, discovery, open-ended lineage.
O-Bench-O0...OOmega	Collective intelligence.	Routing, persistent organization, adaptive coordination, reorganization, meta-improvement, collective discovery, new-role creation.
SA-Bench-SA0...SAOmega	Evidence-grounded self-awareness.	State truth, capability limits, causal self-diagnosis, predicted self-change, self-unknowns, mission identity, lineage.
C-Bench	Broad cognitive capability.	Multiple reasoning/domain families to avoid mistaking one specialty for general cognition.
Circle-Bench	Application/substrate reach, reported orthogonally.	Software closure first; later hardware/physical/biological/other substrate-specific implementation and validation loops.

This organization supports the proposed development workflow: start with Delegation Intelligence because it is immediately operational; then certify a Unified level; then add the coordinate suites most important for the target application. A research worker might emphasize I and SA, a fleet O and DI, an LLM benchmark C and HIL, and a physical autonomy program DI plus Circle completion. The common scores remain comparable because the application panels are reported in addition to, not instead of, the standardized core.

16. Benchmark and Promotion Protocol

- Freeze the tested harness, model, tool set, prompts, and authority profile before certification.
- Use multiple task families per T band so difficulty is not synonymous with one domain such as software.

- Run at predeclared intervention budgets H0-H5 with a written human-assistance policy.
- Use an independent verifier or sealed reference; the performing agent cannot certify its own success.
- Record cognitive interventions separately from governance approvals.
- Record Critical Intervention Depth so a single human-supplied central strategy cannot be hidden by a low intervention count.
- Use confidence intervals and repeated tasks to estimate S_A(T,H), then report the frontier F_A(H,p).
- For U promotion, rerun a retention suite for all lower levels and then run the new-level suite.
- For I3+, O3+, SA4+, and U4+, require longitudinal evidence across multiple self/organizational changes rather than a single successful modification.
- For T5+, use sealed novelty environments, hidden mechanisms, or procedurally generated certification tasks to reduce memorization and benchmark leakage.

17. What Should Remain Outside the Unified Core

Not every important property should be renamed as another intelligence. Several variables are essential for interpreting a system but are better treated as structural, environmental, or substrate coordinates.

Coordinate	**Why it remains outside core intelligence**
Circle / λ	Measures which substrate improvement loops are actually reached or closed; substrate access is not identical to cognition.
Write depth D	Measures which persistent cognitive states can be rewritten; permission to rewrite does not prove intelligent rewriting.
Authority A	Permission is not intelligence. A highly intelligent verifier may deliberately have almost no action authority.
Reach R	Access to files, tools, agents, robots, labs, or networks increases possible action but is not itself cognition.
Verification V	Determines credibility/independence of claims; it validates intelligence rather than constituting the tested intelligence.
Separation S	Organizational validity condition for proposer/evaluator/promoter roles.
Resources	Compute, memory, money, time, energy and hardware affect achievable performance but should not inflate the intelligence definition.
Physical floors φ	Represents irreducible latency/physics constraints and degree of saturation rather than cognitive level.

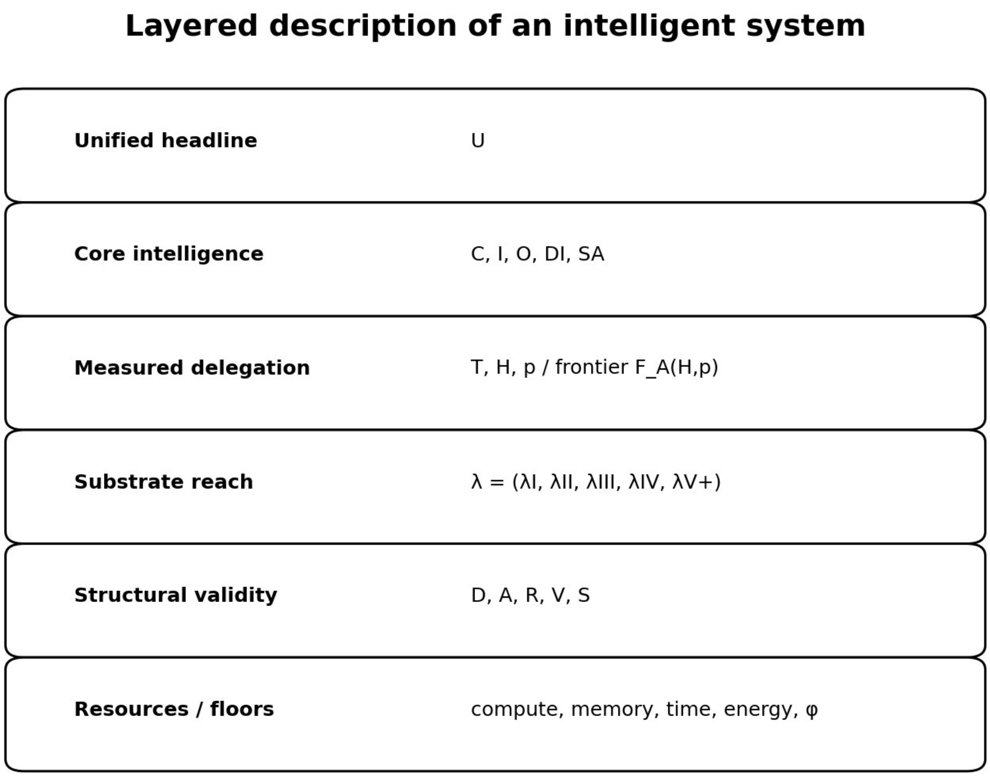

Figure 4. A complete system description layers the Unified headline over the core intelligence profile, measurable delegation frontier, substrate reach, structural validity conditions, and resource/physical constraints.

18. Worked Examples

18.1 Strong LLM, weak harness

[C5, I0, O0, T2, H2, SA1]

The model may solve difficult reasoning problems, but the instantiated agent has little persistent evolution and limited end-to-end delegation autonomy. High C does not automatically produce high U.

18.2 Moderate model, strong persistent harness

[C3, I2, O1, T4, H1, SA2]

This system can be highly useful operationally: persistence, planning, tools, retries, and verification allow difficult tasks to be delegated. Yet it is not self-improving in the I3 sense, and its self-model is not yet causal. The profile communicates these strengths and limits directly.

18.3 Advanced organization with a bottleneck

[C5, I4, O3, T4, H1, SA4] -> U3, bottleneck O3

Several dimensions exceed level three, but organizational intelligence has not reached O4. The unified level remains U3 until the organization demonstrates recursive improvement while retaining lower-level capabilities.

18.4 Same LLM Across a Harness Ladder

A base LLM is tested in HG0, HG1, HG2, and HG3. Its HLIS scores are 42, 55, 67, and 72; the corresponding certified levels are U0, U1, U2, and U3. The model therefore has HIL-Level 3 on the current standardized ladder, a ceiling of 72, and a large positive Harness Gain of 30 points. A second model might start at HLIS 60 in HG0 but rise only to 70 in HG3: it has stronger minimal-harness capability but lower harness gain. HIL-AUC and HIL-Ceiling distinguish these patterns rather than collapsing them into one prompt benchmark.

19. Implications for Intelligence Research

The framework distinguishes model capability from agent capability and both from system capability. A benchmark that tests only answers under carefully prepared prompts primarily measures C. A benchmark that tests whether the same system can accept a difficult goal, maintain state, choose strategy, use tools, recover, verify, and finish with little human cognition measures the delegation surface. Longitudinal tests of learning and self-modification measure I, while multi-agent role and reorganization tests measure O. Grounded introspective tests measure SA.

This decomposition also reduces incentives to overclaim. A system can legitimately report high performance in one dimension without claiming a uniformly high intelligence level. Conversely, the cumulative U hierarchy creates a demanding standard for statements such as 'U4 intelligence': the system must retain lower capabilities and satisfy all required gates at the specified reliability.

The framework is intentionally extensible. Empirical work may revise thresholds or reveal that a currently grouped construct should split. However, new axes should be added only when a property is both operationally measurable and not adequately explained by the existing core or by structural/environmental coordinates.

The HIL framework also creates a new object of study: harnessability. Two LLMs with similar C-scores may differ substantially in their ability to exploit persistent memory, tools, multiple agents, verification loops, or self-improvement infrastructure. A standardized harness ladder can therefore reveal model properties that are invisible in single-turn benchmarks. It also lets researchers quantify whether a new harness architecture produces a general improvement across LLMs or only a model-specific effect.

20. Limitations and Open Questions

- The proposed level thresholds are hypotheses and require empirical calibration across domains and models.
- Task difficulty T is multidimensional; a scalar T band is useful operationally but should retain an underlying difficulty vector for horizon, uncertainty, ambiguity, verification burden, novelty, and environmental change.
- Cognitive Intelligence C may itself require a broad psychometric/domain profile rather than one benchmark family.
- Self-awareness is operational and evidence-grounded; the framework does not address phenomenal consciousness.

- O-levels are only meaningful for organizations with real state/authority/evidence boundaries, not a single prompt that role-plays several agents.
- Open-ended levels require long evaluation horizons and are especially vulnerable to benchmark contamination, resource confounds, and vague success criteria.
- Higher U denotes broader validated capability, not moral value, desirability, safety, speed, or economic efficiency.
- HLIS and HIL weights are provisional. They require calibration against external outcomes, sensitivity analysis, and anti-gaming studies; leaderboard use should publish the full component profile.
- A standardized harness ladder must control tools, context budget, compute, retrieval corpus, authority, and verifier access. Otherwise HIL becomes a measure of infrastructure spending rather than LLM ability.

21. Conclusion

A unified intelligence level is useful only if it summarizes rather than erases structure. This paper therefore proposes five conceptual intelligence families - Cognitive, Individual, Organizational, Delegation, and Self-Awareness Intelligence - while decomposing Delegation Intelligence into two directly measured coordinates, Task Difficulty and Human Cognitive Intervention, conditioned on verified reliability. The resulting scientific profile is [C, I, O, T, H, SA], while the theoretical presentation remains [C, I, O, DI, SA].

Unified Intelligence U0-UΩ is then defined as a cumulative gated hierarchy. Promotion requires retention of all lower-level capabilities and satisfaction of minimum C, I, O, SA and delegation-frontier requirements. This makes it legitimate to say that a higher U-level has a broader validated intelligence envelope, while avoiding the false claim that it must outperform every lower-level system on every narrow metric.

For harness-based AI, the framework provides both a development roadmap and a measurement program. Begin with a minimal LLM harness, add persistence, learning, governed self-improvement, recursive improvement, discovery, organization, mission autonomy, and open-ended transduction. At each stage use a matched hidden dataset. First map Delegation Intelligence over task difficulty and human cognitive intervention; then certify the cumulative Unified level; then add the coordinates most important to the real application. For one concrete LLM-harness pair, report U* and HLIS. For the base LLM, test the standardized harness ladder and report HIL-Level, HIL-AUC, HIL-Ceiling, Harness Gain, and optionally a calibrated HIL-Score. The LLM may even propose better harnesses, but all gains remain subject to frozen external evaluation. This separates what the model can do in isolation, what a harness can unlock, and what the complete intelligent system can actually demonstrate.

Appendix A. Compact Reporting Template

Core measured profile: [C, I, O, T, H, SA] @ reliability p

Conceptual profile: [C, I, O, DI, SA], where DI = F_A(H,p)

Unified headline: U_n [C_x, I_y, O_z, T_k/H_h, SA_s]

Extended system profile: [U; C,I,O,DI,SA; λ; D,A,R,V,S; resources, φ]

Field	Example
Unified headline	U3
Core profile	C5, I4, O3, T4/H1, SA4
Reliability	p = 0.80
Delegation frontier	F_A(H0,.80)=T3; F_A(H1,.80)=T4
Bottleneck	O3
Circle reach	λ = (.42,.08,.03,0,0)
Structural profile	D3, A2, V4, separated proposer/evaluator/promoter
Resource profile	Declared compute, time, memory, energy/cost budget

Appendix B. Suggested Unified Promotion Tests

Transition	Minimum new evidence in addition to full lower-level retention
U0 -> U1	Persistent state and correct goal-level completion after restart or context discontinuity.
U1 -> U2	Verified experience improves future performance on held-out related tasks; multi-step delegation under <=H2.
U2 -> U3	Governed self/organizational method improvement plus T3 strategy construction under <=H1 and causal self-diagnosis.
U3 -> U4	Validated improvement of improvement process plus long-horizon T4 work and accurate modeling of self-change.
U4 -> U5	Independent discovery of initially unknown solution method/knowledge and self-directed experiments about world and self-unknowns.
U5 -> U6	Mission -> projects -> tasks -> execution loop over changing conditions with persistent role/mission self-model.
U6 -> UΩ	Repeated bounded-charter mission generation whose validated discoveries create new tools/agents/organizations that expand future reachable problems.

Framework status. This document is a theoretical proposal developed from the preceding Unified Intelligence, Delegation Intelligence, harness-indexed intelligence, and operational self-awareness design discussion. Level names and numerical thresholds are proposed working definitions and should be revised when benchmark evidence warrants.

Appendix C. Harness-Intelligence Reporting Template

Base LLM: model id/version/hash

Standardized harness ladder tested: HG0, HG1, ..., HGk

Per-harness: U* | HLIS | [C,I,O,T/H,SA] | reliability | bottleneck | resource budget

LLM summary: HIL-Level | HIL-AUC | HIL-Ceiling | Harness Gain | HIL-Score | HDS (if harness-design tasks were run)

Application panel: selected coordinates such as I, O, SA, Circle lambda, verification strength, cost/latency - reported separately from the common core.